GORDON GUOCHENG QIAN

github.com/guochengqian   guochengqian.github.io   (+1) 718 925 1460   guocheng.qian@kaust.edu.sa
485 Ardis Ave, San Jose, CA95117 ⋄ **Interested in Summer Intern 2023**

EDUCATION

King Abdullah University of Science and Technology (KAUST) — Jeddah, Saudi Arabia
P.h.D Candidate in Computer Science — Jan 2021 - Present
Research topics: 3D Perception and Generation, Efficient Neural Networks

King Abdullah University of Science and Technology (KAUST) — Jeddah, Saudi Arabia
Master in Computer Science, GPA: 3.9/4.0 — Aug 2019 - Dec 2020

Xi'an Jiaotong University (XJTU) — Xi'an, China
Bachelor's degree in Engineering with honors, GPA: 3.9/4.0 — Aug 2014 - Jul 2018
Outstanding Undergraduate (top 10 out of 19,000 undergraduates, highest undergraduate honor)
Exchange Student in Japan-Asia Youth Exchange Program in Japan — Feb 2018 - March 2018
Exchange Student at Hong Kong University of Science and Technology (HKUST) — Feb 2017 - May 2017

PUBLICATIONS

I have contributed to 8 peer-reviewed publications in 3D Vision and Efficient Neural Architecture, and most of them have been published in top-tier conferences and journals such as CVPR, NeurIPS, and PAMI. A full list of my publications is available at google scholar: `https://scholar.google.com/citations?user=DUDaxg4AAAAJ`

[1] **Guocheng Qian**, Yuchen Li, Houwen Peng, Jinjie Mai, Hasan Hammoud, Mohamed Elhoseiny, and Bernard Ghanem. **PointNeXt**: Revisiting pointnet++ with improved training and scaling strategies. In *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2022

[2] **Guocheng Qian**, Hasan Hammoud, Guohao Li, Ali Thabet, and Bernard Ghanem. Assanet: An anisotropic separable set abstraction for efficient point cloud representation learning. In *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2021

[3] **Guocheng Qian**, Abdulellah Abualshour, Guohao Li, Ali Thabet, and Bernard Ghanem. Pu-gcn: Point cloud upsampling using graph convolutional networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, June 2021

[4] **Guocheng Qian***, G. Li*, Matthias Müller*, Itzel C. Delgadillo, Abdulellah Abualshour, Ali K. Thabet, and Bernard Ghanem. **DeepGCNs**: Making gcns go as deep as cnns. *IEEE transactions on pattern analysis and machine intelligence (**T-PAMI**)*, 2021

[5] **Guocheng Qian***, G. Li*, Itzel C. Delgadillo*, Matthias Müller, Ali K. Thabet, and Bernard Ghanem. Sgas: Sequential greedy architecture search. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, pages 1617–1627, 2020

[6] **Guocheng Qian**, Yuanhao Wang, Jinjin Gu, Chao Dong, Wolfgang Heidrich, Bernard Ghanem, and Jimmy S Ren. Rethinking learning-based demosaicing, denoising, and super-resolution pipeline. In *IEEE International Conference on Computational Photography (**ICCP**)*, pages 1–12. IEEE, 2022

[7] **Guocheng Qian**, Xuanyang Zhang, Guohao Li, Chen Zhao, Yukang Chen, Xiangyu Zhang, Bernard Ghanem, and Jian Sun. When nas meets trees: An efficient algorithm for neural architecture search. In ***CVPR** Workshop*, pages 2782–2787, 2022

[8] **Guocheng Qian**, Xingdi Zhang, Abdullah Hamdi, and Bernard Ghanem. Pix4point: Image pretrained transformers for 3d point cloud understanding. *arXiv*, 2022

[9] Sicheng Chen, **Guocheng Qian**, Bernard Ghanem, Yongqing Wang, Zhou Shu, Xuefeng Zhao, Lei Yang, Xinqin Liao, and Yuanjin Zheng. Quantitative and real-time evaluation of human respiration signals with a shape-conformal wireless sensing system. ***Advanced Science***, 9(32):2203460, 2022

PROFESSIONAL EXPERIENCE

Meta Platforms Inc Aug 2022 - Feb 2023
- AI Research Scientist Intern, mentored by Dr. Yunyang Xiong
- project: *"M3D: Lightweight Neural Networks for Efficient Video Recognition"*

Megvii Technology Jul 2021 - Nov 2021
- Computer vision intership, directed by Dr. Xiangyu Zhang
- Paper: "When Neural Architecture Search Meets Binary Trees: Remarkable Improvements on Accuracy and Efficiency" (CVPRW'22)

SenseTime Research Aug 2018 - Apr 2019
- Computer vision internship, directed by Dr. Jimmy S. Ren (R&D Director)
- Paper: "Rethinking the Pipeline of Demosaicking, Denoising and Super-Resolution" (ICCP'22)

ACADEMIC EXPERIENCE

Teaching Assistant Deep Learning for Visual Computing (2020 - 2022), Design and Analysis of Algorithms (2021)

Reviewer CVPR, ECCV, NeurIPS, AAAI, T-PAMI, TVCG

Organizer Tutorial on Graph Machine Learning for Computer Vision@CVPR22

HONORS & AWARDS

KAUST CEMSE Research Excellence Award Less than a handful of students are selected.
KAUST CEMSE Dean's List Award Awarded to only 20% of the student population by Division.
KAUST Fellowship Full tuition support, monthly living allowance, housing, and medical coverage.
National Scholarship Top 1%, first-class national scholarship in China.
Outstanding Undergraduate Highest undergraduate honor awarded to 10 selected undergraduates
Excellent Student Cadre Top 3%. Awarded to undergraduate student with leadership

PROGRAMMING SKILLS

- **PyTorch**: PointNeXt >**300 stars**, SGAS >**130 stars**, DeepGCNs >**800 stars**
- **Tensorflow**: PU-GCN (>100 stars)
- **Large scale GPU usage**: multi-node distributed training experience using over 100 GPUs
- **C++, CUDA**: Course projects on CUDA based image processing (), and C++ based curvature estimation ()
- **MATLAB**: experience in image processing, 3d reconstruction (